



DEPARTMENT OF ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

Experiment No.:5

Apply the **K-Means Clustering** algorithm to a dataset containing eight 2-dimensional

Date of Performance:

Particulars	Marks
Regularity(05)	
Performance(15)	
Understanding(Viva)(05)	
Total (25)	
Signature of Course Teacher	



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Aim:

To implement the **K-Means Clustering** algorithm to group the given data points into clusters and update the cluster centroids.

Objectives:

- To understand the concept of clustering in machine learning.
- To apply the K-Means algorithm for grouping similar data points.
- To calculate the distance between data points and centroids.
- To update cluster centroids after assigning points to clusters.

Theory:

Clustering is an unsupervised learning technique used to group similar data points together. One of the most popular clustering algorithms is K-Means Clustering.

In K-Means clustering, the dataset is divided into **K clusters**, where each cluster is represented by a centroid (mean point). The algorithm works by assigning each data point to the nearest centroid and then recalculating the centroid based on the points assigned to that cluster.

The distance between the data point and centroid is usually calculated using Euclidean distance.

The K-Means algorithm repeats the following steps until the centroids stop changing:

- Assign data points to the nearest centroid.
- Recalculate the centroid by taking the mean of the points in the cluster.

This method helps in identifying patterns or groups within the dataset.

Algorithm

1. Select the Number of Clusters (K)

Choose how many groups you want in the dataset.

Example:

$$K = 2$$

So the dataset will be divided into **two clusters (C1 and C2)**.

2. Choose Initial Centroids

Select initial center points for each cluster.

Example:



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$$m1 = P1 = (0.1, 0.6)$$

$$m2 = P8 = (0.3, 0.2)$$

These points act as the starting centers of clusters.

3. Calculate Euclidean Distance

Calculate the distance between each point and the centroids.

Euclidean Distance formula:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

Example for P6 (0.25, 0.5)

Distance from m1:

$$\begin{aligned} d(P6, m1) &= \sqrt{[(0.25 - 0.1)^2 + (0.5 - 0.6)^2]} \\ &= \sqrt{(0.0225 + 0.01)} \\ &\approx 0.18 \end{aligned}$$

Distance from m2:

$$\begin{aligned} d(P6, m2) &= \sqrt{[(0.25 - 0.3)^2 + (0.5 - 0.2)^2]} \\ &= \sqrt{(0.0025 + 0.09)} \\ &\approx 0.30 \end{aligned}$$

Since $0.18 < 0.30$, P6 is closer to m1.

So P6 belongs to Cluster C1.

4. Assign Points to Nearest Centroid

After calculating distances, assign each point to the closest centroid.

Example cluster formation:

Cluster C1

P1, P2, P3, P4, P6

Cluster C2

P5, P7, P8

5. Recalculate the Centroids

Compute the new centroid by taking the average of all points in each cluster.

Example for Cluster C1

Points:

P1(0.1, 0.6)



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P2(0.15,0.71)

P3(0.08,0.9)

P4(0.16,0.85)

P6(0.25,0.5)

New centroid m1:

$$x = (0.1 + 0.15 + 0.08 + 0.16 + 0.25) / 5 \\ = 0.148$$

$$y = (0.6 + 0.71 + 0.9 + 0.85 + 0.5) / 5 \\ = 0.712$$

Updated m1 = (0.148 , 0.712)

6.Repeat Until Centroids Do Not Change

The algorithm repeats the steps:

1. Calculate distance
2. Assign clusters
3. Update centroids

until the centroid values stop changing.

At that point, the clusters are considered stable.

Conclusion:

In this experiment, the K-Means Clustering algorithm was applied to group the given data points into two clusters using Euclidean distance. The points were assigned to the nearest centroid and the centroids were updated by calculating the mean of the cluster points. This experiment helps to understand how K-Means clustering organizes data into meaningful groups.